SIMATS ENGINEERING
ASSESSMENT TOOL 2

COURSE NAME: CSA07 COURSE CODE: Computer Networks

COURSE OUTCOME COVERED

CO1: *Apply the functions of OSI layer in enabling reliable data transmission across networks. (BL2, BL3)*

Assessment Tool Weightage: 30%

Annexure A

CONCEPT MAPPING

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Code of Conduct:

I Jeevan Raaja S (192421005) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment.
I understand that any violation of academic integrity rules will result in disciplinary action.

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Computer Network

CONCEPT MAPPING – Questions with Answers (OSI Model)

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- **A partially completed concept map is given below:**

Question:

Application → Presentation → ________→ Transport → Network → _______→ Physical

Analyze the missing layers and explain how their functions are essential for reliable communication.

Answer:

Completed Concept Map

Application
↓
Presentation
↓
Session
↓

Transport

↓

Network

↓

Data Link

↓

Physical

Missing Layers

Session Layer Data

Link Layer

Explanation

Session Layer (Layer 5)

Establishes communication sessions between devices. Maintains and synchronizes communication.

Terminates the session after communication is completed.

Supports dialog control and synchronization.

Data Link Layer (Layer 2)

Transfers data between directly connected devices. Detects and corrects transmission errors.

Uses **MAC addresses** for local delivery. Converts packets into frames.

Importance for Reliable Communication

The **Session Layer** ensures continuous and organized communication.

The **Data Link Layer** provides reliable local transmission through framing and error detection.

- **Construct a concept map linking data units (Segment, Packet, Frame, Bits) with their respective OSI layers and analyze how transformation of data units supports transmission.**

Answer Concept

Map

```
Application
   ↓
Presentation
   ↓
Session
      ↓
   Transport


   |
   └── Segment
        ↓
   Network
      |
      └── Packet
           ↓
   Data Link
      |
      └── Frame
           ↓
```

```
Physical
   |
   └── Bits
```

Data Units

OSI Layer	Data Unit
Transport	Segment
Network	Packet
Data Link	Frame
Physical	Bits

Analysis

Transport Layer divides data into **segments**. **Network Layer** adds IP information to create **packets**.

Data Link Layer adds MAC address and error-checking to create **frames**.

Physical Layer converts frames into **bits** for transmission.

This process is called **Encapsulation**.

At the receiver, the reverse process (**Decapsulation**) reconstructs the original data.

- **Given the concept map:**

Data Link → MAC Address → Local Delivery

Network → IP Address → Global Delivery

Explain how these relationships ensure complete data delivery across networks.
Answer Concept Map

Data Link Layer

↓

MAC Address

↓

Local Delivery

Network Layer

↓

IP Address

↓

Global Delivery

Explanation Data Link

Layer

Uses **MAC addresses**.

Delivers frames between devices in the same LAN. Responsible

for local communication.

Network Layer

Uses **IP addresses**.

Routes packets across different networks.

Responsible for global communication.

Analysis

Together,

IP addresses determine the destination network.

MAC addresses deliver the data to the correct device within that network.

Hence both layers work together for complete end-to-end communication.

- **A concept map shows:**

Error Control

→ Data Link Layer → Error Detection

→ Transport Layer → Error Correction

Analyze how the interaction between these layers improves reliability in data transmission.

Answer Concept

Map

```
        Error Control
         /        \
        /          \
Data Link Layer   Transport Layer
     ↓                ↓
Error Detection   Error Correction
```

Explanation Data Link

Layer

Detects transmission errors using:

CRC (Cyclic Redundancy Check) Checksum

Parity bits

○

Transport Layer

Provides reliable communication using:

Acknowledgements (ACK) Retransmission

Sequence numbers Flow

control

Analysis

The Data Link Layer identifies transmission errors, while the Transport Layer ensures lost or corrupted data is retransmitted. Together, they provide reliable end-to-end communication.

- **Develop a concept map for web browsing using OSI layers and analyze how failure in one layer affects the entire communication process.**

Answer Concept

Map

Web Browser

↓

Application Layer (HTTP)

↓

Presentation Layer (SSL/TLS)

↓

Session Layer

↓

Transport Layer (TCP)

↓

Network Layer (IP)

↓

Data Link Layer (Ethernet/Wi-Fi)

↓
Physical Layer (Cable/Wireless)

Explanation

Layer	Function
Application	HTTP request
Presentation	Encryption (SSL/TLS)
Session	Session management
Transport	TCP reliable delivery
Network	Routing using IP
Data Link	Local delivery using MAC
Physical	Transmission through cables or wireless

Analysis

Failure at any layer affects communication. Examples:

Physical failure → No signal.

Data Link failure → Frames cannot be delivered.

Network failure → Routing stops. Transport

failure → TCP connection fails.

Application failure → Website cannot be displayed.

Thus, every OSI layer depends on the services of the layer below it.

- **A network issue is represented in the concept map:**

Physical ✓

↓

Data Link ✓

↓

Network ✕

↓

Transport ✕

↓

Application ✕

Analyze the problem and explain how the concept map helps in troubleshooting.

Answer Concept

Map

Physical Layer ✓

↓

Data Link Layer ✓

↓

Network Layer ✕

↓

Transport Layer ✕

↓

Application Layer ✕

Explanation

Since:

Physical Layer is working Data Link

Layer is working

The problem starts at the **Network Layer**. Possible

causes:

Incorrect IP address Wrong

subnet mask Missing default

gateway

Routing failure Router

malfunction

Because the Network Layer fails:

Transport Layer cannot establish TCP/UDP communication.

Application Layer cannot access services such as web browsing or email.

Troubleshooting Steps Check IP

configuration. Verify subnet

mask.

Check default gateway.

Test router connectivity using **ping**. Verify

routing tables.

Check DNS configuration if routing is working.

Analysis

The concept map clearly identifies that the fault begins at the **Network Layer**, making troubleshooting easier by narrowing the problem to routing or IP configuration rather than physical connectivity.

Summary

OSI Layers: Application → Presentation → Session → Transport → Network → Data Link → Physical

Data Units: Segment → Packet → Frame → Bits

Addresses: MAC Address (Local Delivery), IP Address (Global Delivery)

Reliability: Data Link detects errors; Transport corrects errors and ensures reliable delivery.